

EDUCATION

Georgia Institute of Technology

Bachelor of Science in Mechanical Engineering (Manufacturing Concentration)

Atlanta, GA

Aug. 2023 – May 2027

Georgia Tech Europe (GTE)

Study Abroad Program

Metz, France

Jan. 2025 - May 2025

EXPERIENCE

Smurfit Westrock

Process Engineering Intern

May 2025 – Aug. 2025

Mooreville, NC

- Led process engineering analysis across corrugated manufacturing systems (thermal, mechanical, quality) projecting \$275K in annual savings through waste reduction and equipment utilization.
- Directed development of a data-driven inventory optimization system (Oracle JD Edwards API integration) to support continuous manufacturing operations, reducing excess stock in a \$13.52M inventory by 23%.
- Partnered with maintenance technicians on equipment diagnostics and inventory control initiatives, integrating lean manufacturing and 5S practices to improve machine uptime.

Snapping Shoals EMC

Automotive Intern

Aug. 2022 – May 2023

Covington, GA

- Performed preventive maintenance on a fleet of ~75 utility vehicles including oil/filter service, brake inspections, electrical diagnostics, and hydraulic checks, supporting operations for a \$12M asset base.
- Worked alongside shop mechanics on chassis and aerial-lift system improvements (hydraulic inspection, chassis integrity assessment, outrigger and boom lubrication) to improve serviceability and assembly efficiency.
- Executed paint correction, multi-stage bodywork (bare-metal prep, filler work, sealant application), surface refinishing, and on-board diagnostics (OBD fault-code analysis, sensor verification).

Curious Cardinal

Mentor

Dec. 2024 – Present

Atlanta, GA

- Mentored K-12 students in personalized one-on-one sessions, designing customized curricula in STEM and arts disciplines.
- Coached students through professional development and career guidance, including college application processes and the creation of research and creative portfolios.

PROJECTS

Autonomous Robot - ME 2110 | CAD, Mechatronics, Prototyping and Fabrication, DFM, FMEA

Aug. 2025 – Dec. 2025

- Designed, fabricated, and programmed an autonomous robot for Georgia Tech's ME 2110: Creative Decisions & Design competition, placing 2nd overall in a class of 70+ teams; integrating Arduino-based mechatronic control and custom modular subsystems to complete five mechanical tasks in under 40 seconds.
- Implemented mechanical design and manufacturing principles, including DFM/DFA, FMEA, and iterative prototyping to optimize subsystem reliability and assembly efficiency, utilizing CAD, digital fabrication, 3D printing, and electrical/mechanical tools.

R&D: RV-14 Aircraft (Yellow Jacket Aviation) | Process & Build Sequencing, MFG & Process Coordination

May 2025 – Present

- Developed and owned an early-stage fabrication readiness plan for the RV-14 aircraft, defining part sourcing, BOM, kitting approach, and initial build sequencing to support the launch of assembly and prevent part shortages or build interruptions.
- Coordinated cross-functional teams (structures, systems, and logistics) during the start of assembly to track part readiness, standardize initial build procedures, and resolve early supply and process constraints, improving build flow and reducing rework risk.

INVOLVEMENT AND LEADERSHIP

Yellow Jacket Aviation

Research Team Lead

Feb. 2023 – Present

- Lead of a YJA's 25-member research team, spearheading R&D of an FAA-compliant aircraft, coordinating interdisciplinary engineering efforts and collaborating with logistics and design teams for a legal, manufacturable, and cost-effective strategy.

American Society of Mechanical Engineers

General Body Member

Sep. 2023 – Present

- Participate in weekly professional development events and networking sessions with top-tier engineering firms to enhance industry knowledge and career opportunities.

TECHNICAL SKILLS

CAD: Solidworks, Fusion 360, Onshape, PTC Creo, Autodesk Inventor, CATIA

Engineering: Finite Element Analysis (FEA), Data Analytics (Matlab, Python, Excel, Power BI), GD&T, Prototyping and Iteration, 3D Printing (FDM), Arduino/Microcontroller Programming, Robotic System Integration, Digital Fabrication, Controls and Actuation

Relevant Coursework: Thermodynamics, Dynamics of Rigid Bodies, Mechanics of Materials, Mechanics of Deformable Bodies, ECE

Manufacturing: Lean Manufacturing (5S, Six Sigma, Kaizen, Standard Work), Process Improvement and OEE, RCA & Defect Reduction, SOP Development, Quality Control, Material Characterization, Corrugated Machinery

Certifications: Six Sigma Yellow Belt, CompTIA Network+, NOCTI Security